

A Dam Catch-22.3: The Non-Linear Spatial Effects of North Carolina's Water Infrastructure on Housing Markets*

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Abstract

Infrastructure projects generate complex spatial externalities that ripple through local housing markets. These factors influence home values in ways which traditional real estate markets sometimes fail to capture. In this paper, I utilize cross-sectional data variation in dam locations across 747 Zip Code Tabulation Areas (ZCTAs) in North Carolina to estimate the capitalization of dam proximity into housing values. Theoretical models suggest a non-linear tradeoff where reservoirs act as recreational amenities, while dam structures pose disamenity risks. Using a hedonic pricing model with quadratic functional forms, I initially find modest relationships in standard Ordinary Least Squares (OLS) estimates. However, after correcting for the endogeneity of residential sorting via an Instrumental Variable (IV) approach, I find a statistically significant non-linear relationship ($p < 0.05$). The results indicate that dam proximity serves as a net amenity for approximately 22.3 miles before the effect dissipates. These findings suggest that the amenity capitalization of water infrastructure is spatially extensive and robust to controls for structural quality and population density.

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I. Introduction

The valuation of environmental amenities remains a central challenge in urban economics. Large scale infrastructure projects, such as dams and reservoirs, create a unique dual-nature externality: they provide recreational utility and aesthetic value (an amenity) while simultaneously introducing flood risk and visual obstruction (a disamenity). Understanding how these opposing forces capitalize into housing markets is critical for cost-benefit analysis, urban planning, and tax assessment (Bohlen & Lewis, 2009).

This paper investigates the spatial relationship between dam proximity and residential property values in North Carolina. Utilizing a comprehensive dataset of 747 Zip Code Tabulation Areas (ZCTAs) and geolocated data for over 3,600 dams, I employ a Hedonic Pricing Model to isolate the implicit price of distance. Unlike previous studies which often assume linear relationships, or perhaps altogether ignore the complex dynamics of residential choice, this analysis explicitly tests for curvilinear gradients and corrects for the endogeneity of population density using an instrumental variable strategy.

My analysis yields two primary findings. First, I demonstrate that standard Ordinary Least Squares (OLS) estimation suffers from attenuation bias due to the endogenous sorting of high-density populations into amenity-rich areas—a phenomenon consistent with Tiebout sorting (Tiebout, 1956). The significance of the endogeneity test ($p < 0.01$) confirms the necessity of this strategy, revealing a statistically significant non-linear relationship between dam proximity and housing values that was otherwise obscured. Housing values increase as distance from the dam increases, but this growth diminishes over space, reaching an inflection point at approximately 22.3 miles. This confirms the hypothesis that reservoirs (dams) act as regional amenities that anchor property value, with influence that extends well beyond the immediate perimeter (Loomis & Feldman, 2003).

The remainder of this paper is organized as follows. Section II presents the current literature that is pertinent to the analysis performed during research and also reviews the theoretical framework of hedonic pricing and its adjacent economic corollaries. Section III details the data sources and econometric methodology, specifically the construction of the instrumental variable. Section IV presents the

empirical results while discussing both the structural estimates and the spatial amenity gradient. Section V concludes with a summation of findings as well as how these results impact current economic frameworks and policy implications and provides potential avenues for future research. A list of references for works cited throughout the paper is provided as well as an “Additional Figures and Tables” section that provides supplementary graphics and illustrations, produced in my analysis, and serves to further support results. Finally, the appendix portion lists snippets of code used in conducting specific elements of data refinement and regression analysis for the purpose of study replication.

II. Literature Review and Theoretical Framework

A. Related Literature

While literature on environmental capitalization is extensive, it typically focuses on single-direction externalities. The term “single direction” is used to describe the pricing effect an externality imposes on a good or service, as uniquely positive or negative. In other words, hedonic analysis has predominately examined externalities as either exclusively increasing the value of a good/service or exclusively decreasing the value. For example, Bui and Mayer (2003) demonstrate the negative capitalization of toxic release sites, while other studies consistently find positive capitalization from coastal amenities. Inland infrastructure such as dams, however, present an opportunity to examine this framework under an entirely new application of hedonic analysis. Human made dams provide the necessary characteristics to study this theoretical puzzle: they function simultaneously as a recreational amenity (reservoirs, recreation activities, water utility, etc.) and a structural disamenity (flood risk, visual blight, noise pollution, etc.). Current research utilizing Census-level data often struggles to capture structural quality of homes as predictors in housing value. This study utilizes median room counts as a rigorous proxy for structural capacity to mitigate this omitted variable bias (Gyourko et al., 2013).

B. Hedonic Pricing Method

This paper’s analysis builds upon the foundational Hedonic Pricing Model established by Rosen (1974), which conceptualizes housing not as a homogeneous good, but as a bundle of structural, neighborhood, and environmental characteristics. It states that a differentiated good, such as housing, can be viewed as a bundle of attributes $(z_1, z_2, \dots, z_n)$. In a competitive equilibrium, the price of the house (P) is a function of these attributes. The market price of a house represents the sum of the implicit prices of these attributes such that:

$$P(z) = f(S, N, L) \quad (1)$$

Where S represents structural characteristics (e.g., room count), N represents neighborhood characteristics (e.g., density, income), and L represents locational amenities (e.g., distance to infrastructure). Incorporating traditional Hedonic framework into my specific use case to model the capitalization of dams, I propose a “Dual-Nature” spatial framework. Unlike traditional economic literature which understands a standard linear amenity as “closer is better”, the net price effect of a dam in this examination is hypothesized to be the summation of two opposing distance-decay functions:

1. The Amenity Function: A positive premium derived from reservoir views and recreational access, which decays gradually over distance.
2. The Disamenity Function: A negative discount derived from flood risk and infrastructure proximity, which decays rapidly immediately surrounding the structure.

C. Theory of Amenities

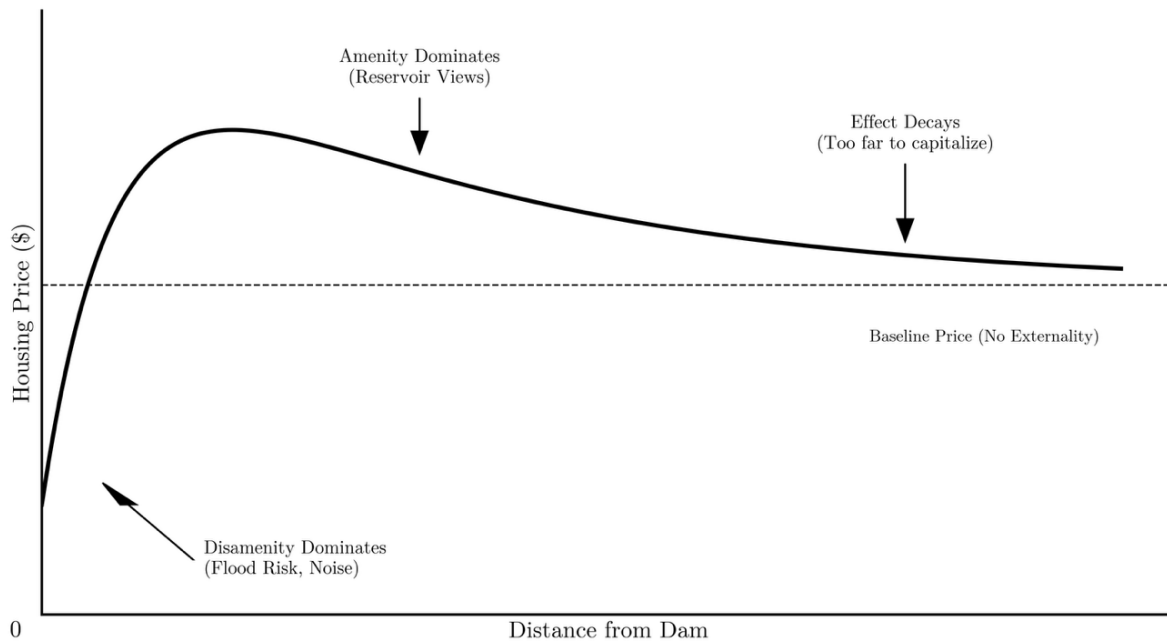
Classic urban economic theory (Alonso, 1964) suggests that housing prices decay with distance from a central business district (CBD). However, environmental amenities introduce local peaks in this price gradient. As illustrated in Figure 1, the theoretical expectation for dam proximity is a non-monotonic curve.

- Disamenity Zone ($d \approx 0$): Properties immediately adjacent to the dam structure may suffer from visual blight, noise, or perceived risk, depressing values.
- Amenity Zone ($d > 0$): Moving slightly away, the recreational benefits of the reservoir (boating, views) dominate, raising property values.
- Distance Decay: As distance (d) increases further, the amenity value fades to zero, and prices stabilize at the regional baseline.

This study empirically tests the existence of this “Amenity Zone” by employing a quadratic functional form (Distance + Distance²) to capture the inflection point where the marginal implicit price of distance shifts from negative to positive.

Figure 1’s* graph depicts a theoretical “U-shape” or “S-curve” where the net effect transitions from negative to positive before decaying to zero.

Figure 1: Theoretical Capitalization of Dam Proximity



* Figure 1 illustrates theoretical tension; prices are suppressed in the immediate vicinity of the structure (Disamenity Dominance) but rise to a premium at a moderate distance where views are maximized (Amenity Dominance), before decaying to the baseline. The point where the net price curve intersects the baseline represents the theoretical breakeven distance. At this specific location, the decaying positive amenity of the reservoir view is perfectly offset by the decaying negative disamenity of the dam structure, resulting in a net capitalization effect of zero.

The theoretical prediction relies on the assumption that disamenity decays faster than the amenity, which generates the non-linear price gradient. Flood risk and noise are localized to the immediate and surrounding vicinities of the dam, whereas recreational access and beautiful views provide value (utility) over a wider geographic radius. The simultaneous presence of these amenities is at the core of my study. As these attributes continuously produce contradicting effects on surrounding towns and communities, the application of hedonic pricing theory becomes crucial in understanding the quantitative outcomes on housing prices.

D. Geographic Background

North Carolina offers a unique setting to test this framework due to its dense network of regulated dams and distinct hydrology (Kastrinsky, 2022). Unlike the arid west where water is exclusively a premium, North Carolina's hydrology introduces substantial flood management risks downstream of these structures. By analyzing this relationship at the Zip Code Tabulation Area (ZCTA) level, this study examines whether these localized amenity-risk tradeoffs are sufficiently powerful to capitalize into aggregate property values, or if they are washed out by broader economic fundamentals.

III. Data and Methodology

Seeking to understand the non-linear capitalization of reservoir (dam) amenities and flood risk disamenities into local housing markets, a theoretical equation was formally derived in Equation (1) (see Section II). To empirically test the predicted tradeoff, it was necessary to determine data availability and compatibility across specified geographic regions. The construction of a unified, cross-sectional dataset at the Zip Code Tabulation Area (ZCTA) level proved to be the most accurate and descriptive data available for testing my hypothesis.

I sourced and synthesized publicly available data from the United States Census Bureau, the United States Army Corps of Engineers (USACE), the Department of Environmental Quality for North Carolina (DEQ) and the U.S. Census Gazetteer files to build the data set necessary to conduct this study. The dependent variable, median housing value, is represented by the median home value of a Zip Code

Tabulation Area (ZCTA). The United States Census Bureau publishes socioeconomic data at the ZCTA level through the American Community Survey (ACS). Because housing markets are highly localized, I selected the ZCTA level as my study geography for the purpose of capturing variation that might be smoothed over at the county level. It should be noted that throughout the remainder of this paper, the variables Distance and Nearest Dam Distance are used interchangeably as they measure identical quantities within the parameters of my analysis.

Table 1: Data Description, Units and Sources

Variable	Description	Units	Source
Log Median Home Value	Log of median home value (ZCTA)	Dollars	US Census (ACS 2023)
Nearest Dam Dist	Geodesic distance to nearest dam	Miles	Calculated (Census + National Dams)
Distance Squared	Quadratic distance term	Miles ²	Calculated
Log Median Income	Log of median household income	Dollars	US Census (ACS 2023)
Log Pop. Density	Log of population per sq mile	People	Calculated (Census + Gazetteer)
Log Median Rooms	Log of median room count	Rooms	US Census (ACS 2023)

A. Variable Definition and Transformation

All socioeconomic data was obtained from the American Community Survey (ACS) 5-Year Estimates through a data filtering tool provided by the U.S. Census Bureau on their website. These variables were Median Home Values, Median Household Income, Median Rooms, and Population Density for all North Carolina ZCTAs ($N = 747$). To ensure analytical rigor and comparability across the sample, all monetary variables were calculated in 2023 inflation-adjusted dollars. I performed specific data cleaning and transformations to normalize the distribution of skewed economic variables. The National Inventory of Dams (NID) also provided precise coordinates for 3,641 dams within the state. Further detail is provided below regarding the different variable distinctions used in the regression models.

- **Dependent Variable:** I defined the dependent variable using the Median Home Value from the ACS. Prior to transformation, I identified records containing non-numeric string identifiers (such as “-” or “N/A”)

instead of values. I changed these records to null values and removed them from the dataset to prevent numerical errors and ensure all observations contained attributes relevant to the regression analysis. To account for inflation, the U.S. Census Bureau adjusts all figures to 2023 inflation-adjusted dollars. This allowed for direct comparisons and seamless integration with the regression models. After collection and removal, the remaining values were transformed into their natural logarithm to create the variable `Log_Home_Value`.

- **Distance Metrics:** To measure the proximity to infrastructure, I sourced the location of all dams in North Carolina from the National Inventory of Dams (NID). Using the *geopy* Python library, I calculated the distance for each ZCTA centroid to the closest dam using a k-d tree algorithm and geodesic measurements. To test for the non-linear amenity effects predicted in my theoretical framework, I also calculated the square of this distance, and produced the variable `Distance Squared`.
- **Control Variables:** I defined several covariates using data gathered from the Census Bureau. Median Household Income serves as a control for the purchasing power of consumers/households within the geographic statistical areas (ZCTAs). Similar to the dependent variable, this was adjusted to 2023 dollars and log-transformed to interpret the coefficient as an elasticity. Median Household Rooms was included to address potential omitted variable bias regarding the structural characteristics of the housing stock. It also served as a proxy for household square footage. Although the dimensions (square footage) of a house are most likely a stronger predictor of price, the U.S. Census does not capture that information through their surveys. Household Median Rooms is instead the standard academic proxy for size in Census-based research—it is understood as the best volume metric available for addressing structural influence on price. Finally, Population Density was calculated by dividing a ZCTA's total population by its land area (in square miles) as reported in the U.S. Gazetteer files.

Table 2: Descriptive Statistics

Variable	Units	Mean	Std. Dev.	Min	Max
Log Median Home Value	Dollars	12.2339	0.5160	10.5321	14.1148
Nearest Dam Dist	Miles	5.1715	10.2941	0.0428	74.5916
Distance Squared	Miles^2	132.5715	591.6082	0.0018	5563.9093
Log Median Income	Dollars	11.0493	0.3270	9.8932	12.2660
Log Pop. Density	People	5.1372	1.4811	0.4450	9.1422
Log Median Rooms	Rooms	1.7202	0.1087	1.1632	2.1401

As briefly stated above, simple data transformations, commonly utilized throughout economic studies concerning housing data, were implemented for a portion of the variables to provide more meaningful results. Variables including home value, household income, and population density were transformed into their natural logarithms to normalize their naturally right-skewed distributions and mitigate the influence of extreme outliers common in wealth and geospatial data. Furthermore, this log-linear specification allows for the interpretation of coefficients as elasticities, effectively capturing the percentage change in housing values associated with relative changes in socioeconomic conditions. Finally, this functional form accounts for diminishing marginal returns in structural attributes, reflecting the economic reality that an additional unit of amenity (such as room count) yields smaller value premiums as the total quantity increases. The logarithmic transformation of Population Density was particularly critical given the extreme spatial heterogeneity of North Carolina, which encompasses both sparse agricultural ZCTAs and highly dense metropolitan centers like Charlotte. A linear specification would allow these high-density outliers to disproportionately influence the regression estimates. Furthermore, because population density is non-linearly determined by land area, ($\text{Density} = \text{Population} / \text{Land Area}$), log-transforming both variables ensures a linear first stage relationship between the endogenous variable and the instrument, Log_Land_Area , thereby satisfying the functional form requirements of the identification strategy.

The correlation matrix confirms the presence of structural collinearity necessary for the model's functional form. Specifically, the relationship between Distance and Distance², which is an expected mathematical outcome of employing a quadratic

specification to capture the non-linear amenity/disamenity tradeoff. While this near-perfect collinearity renders the individual t-statistics for these two variables unreliable, it does not invalidate the model. Following established econometric procedure, I successfully assessed the joint significance of these two terms using the F-Test, which confirmed their collective importance in the final IV model. Figure 2 confirms that multicollinearity among controls remains within acceptable limits for the remaining variables within a hedonic model (Anselin and Griffith, 1988).

Figure 2: Correlation Between Variables



B. Econometric Specification

I estimate three specifications of the hedonic price function to rigorously test the capitalization of dam proximity while controlling for structural and spatial heterogeneity.

To establish a baseline, I employ a semi-logarithmic Ordinary Least Squares (OLS) specification. The semi-log form is standard in environmental hedonic literature as it interprets coefficients as semi-elasticities (percentage changes in price). This functional form also accounts for the non-linear relationship typically observed between housing values and density. The benchmark estimating equation is defined as equation 2:

$$\begin{aligned} \text{Log_Home_Value} = & \beta_0 + \beta_1 \text{Distance} + \beta_2 \text{Distance}^2 \\ & + \beta_3 \text{Log_Median_Income} \\ & + \beta_4 \text{Log_Pop_Density} + u \end{aligned} \quad (2)$$

In this specification, the primary coefficients of interest are β_1 and β_2 . The inclusion of the quadratic term (Distance^2) is critical to testing the non-monotonic hypothesis. A statistically significant coefficient for the squared term would confirm that the amenity effect is not linear but rather decays or reverses over space, allowing for the calculation of a theoretical turning point where the net amenity value dissipates.

A primary threat to identification in spatial hedonic models is Omitted Variable Bias (OVB). It is plausible that larger, higher quality housing stock systematically sorts into specific locations relative to water infrastructure (e.g., larger properties near reservoirs or rural lots further away). If left uncontrolled, the distance coefficient (β_1) would capture the premium of the structure rather than the location.

Model 2 addresses this by expanding the vector of controls to include `Log_Median_Rooms`. By holding the median size of the housing price constant, this specification isolates the amenity value of the dam from the physical characteristics of the homes in the ZCTA. The OLS estimates in Models 1 and 2 rely on the assumption that Population Density is exogenous. However, contemporary urban economic theory suggests this variable is likely endogenous due to simultaneity bias: high amenity areas naturally attract residents which leads to increasing density, and in turn influences prices (Lee & Lin, 2015).

Equation 3 (Control Function / IV) Addresses the endogeneity of Population Density. High amenity areas may attract higher density, which in turn affects price (Simultaneity Bias). Because high price (or the unobserved factors driving high price, captured in the error term u) directly affects Population Density, the

coefficient on Population Density becomes biased. The OLS estimate will capture not just the structural effect of density, but also the reverse causal flow, leading to an overestimation of the coefficient. I utilize *Log_Land_Area* as an instrument for density. This IV works because it meets the two necessary requirements for a valid instrument:

1) *Log_Land_Area* is highly correlated with Density (Population Density = Population/Area). ZCTAs with larger amounts of land area reduce population density (assuming population is held constant) and ensures the instrument is a good predictor of the endogenous variable.

2) The size of the land area of the ZCTA is determined by administrative or geographical boundaries and is not directly correlated with the unobserved amenities (such as reservoir views) that drive housing prices. Therefore, *Log_Land_Area* is exogenous to the error term u and can isolate the exogenous variation in Population Density and obtain an unbiased estimate of its true effect on *Log_Home_Value*. The first stage residual is included in the second stage regression to explicitly test and correct for this bias.

The 2SLS Control Function (IV) strategy proceeds in two stages. First, the endogenous variable (*Log_Population_Density*) is regressed on the instrument (*Log_Land_Area*) and all other exogenous covariates (distance, income, rooms).

$$\begin{aligned} \text{Log_Pop_Density} = & \pi_0 + \pi_1 \text{Log_Land_Area} + \pi_2 \text{Distance} + \pi_3 \text{Distance}^2 \\ & + \pi_4 \text{Log_Median_Income} \\ & + \pi_5 \text{Log_Median_Rooms} + v \end{aligned} \quad (3)$$

The residual from this first stage is included as an additional regressor in the main hedonic price equation (2). The statistical significance of this residual term in the second stage serves as a direct test for endogeneity. The statistical significance indicates that the unobserved factors influencing density also influence price, which require the IV correction to obtain unbiased amenity estimates.

IV. Results and Discussion

A. Structural Estimates and Controls

Across all models, the control variables behave consistently with economic theory regarding home values and the predominant variables that influence price. The R-squared values for each Model are 0.618, 0.622 and 0.631 respectively. This growth across models indicates that each change in composition positively affected the explanation of variation in the dependent variable, signifying the decisions to incorporate Log_Median_Rooms and introduce an Instrument Variable were justified. The control function, specifically, has the highest of the three reported R-squared values where 63.1% of the variation in Log_Home_Value is explained by the model. It is noteworthy that Log_Median_Income is the strongest predictor of home value ($\beta \approx 1.23$, $p < 0.01$ as produced by Model 3), confirming that neighborhood socioeconomic status is the primary driver of price. This is consistent with hedonic economic theory on housing values. Log_Median_Rooms (Model 2 & 3) is statistically significant ($p < 0.01$ on both models), validating its inclusion to address the potential Omitted Variable Bias (OVB).

Taking a deeper look into Model 3 and the introduction of the Instrument Variable is essential in understanding the implications for attributing new causal evidence to the analysis which didn't exist during the regression of Models 1 and 2. The first stage of the 2SLS process utilized Log_Land_Area as an Instrumental Variable (IV) to isolate the variation in Log_Population_Density that is truly exogenous to housing price. By including the residual from this first stage (First_Stage_Resid.) in the final regression (Model 3), we simultaneously test for and correct for endogeneity. The highly significant positive coefficient on the First_Stage_Resid. ($\beta = 0.0899$, $p < 0.01$) is the definitive evidence that endogeneity was present (rejecting the null hypothesis).

The final coefficient of interest, Log_Population_Density, is then interpreted as the true and unbiased causal effect. In Model 3, this coefficient drops sharply to 0.0067 and becomes statistically insignificant ($p=0.819$). It is imperative to appreciate this result as a profound change from the biased OLS estimate of 0.0760***. Such divergence confirms the OLS model was severely overestimating the direct effect of population density, effectively masking the true and

significant non-linear implications of dam proximity that was revealed only after correcting for a residential sorting bias.

My analysis confirms that residential sorting introduces significant bias into standard hedonic models. This result implies that unobserved factors driving density—such as neighborhood prestige or local public goods—are correlated with the error term in the price equation. By instrumenting density with land area, I isolate the variation in density that is independent of these unobserved price shocks. Correcting for this endogeneity bias was crucial as it revealed a statistically significant amenity premium for dam proximity ($p < 0.05$) that was masked in the biased OLS specifications.

B. The Amenity Effect (Model 3)

The regression results are summarized in Table 3 (following page), where the most critical finding emerges in Model 3. In the OLS specifications (Models 1 & 2), the linear distance term is statistically insignificant or weak. However, in the Control Function specification (Model 3), the linear term becomes negative ($\beta_1 = -0.0134$) and statistically significant ($p < 0.05$). This is aligned with the theoretical projections for the effects of dam infrastructure on housing prices. The negative linear coefficient indicates that, holding all else constant, moving away from a dam reduces home value. This confirms the dam/reservoir acts as an amenity. The positive squared term ($\beta_2 = 0.0003$) is significant ($p < 0.01$), indicating that this negative price gradient flattens out as distance increases.

Table 3: OLS and Instrument Variable Regression Results*

Variable	Model 1 (OLS)	Model 2 (OLS)	Model 3 (IV)
Constant	0.6680 (0.4636)	0.5757 (0.4985)	-0.2769 (0.5655)
Nearest Dam Dist (mi)	-0.0053 (0.0045)	-0.0042 (0.0045)	-0.0134** (0.0057)
Distance Squared	0.0002*** (0.0001)	0.0002** (0.0001)	0.0003*** (0.0001)
Log Median Income	1.0081*** (0.0435)	1.0925*** (0.0593)	1.2322*** (0.0750)
Log Pop. Density	0.0824*** (0.0105)	0.0760*** (0.0110)	0.0067 (0.0293)
Log Median Rooms		-0.4709*** (0.1489)	-0.6483*** (0.1587)
1st Stage Resid. (Endogeneity Test)			0.0899*** (0.0346)
Observations	749	747	747
R-squared	0.618	0.622	0.631

Using the coefficients from the preferred Model 3, I calculate the spatial extent of this amenity effect using the turning point formula for a quadratic function:

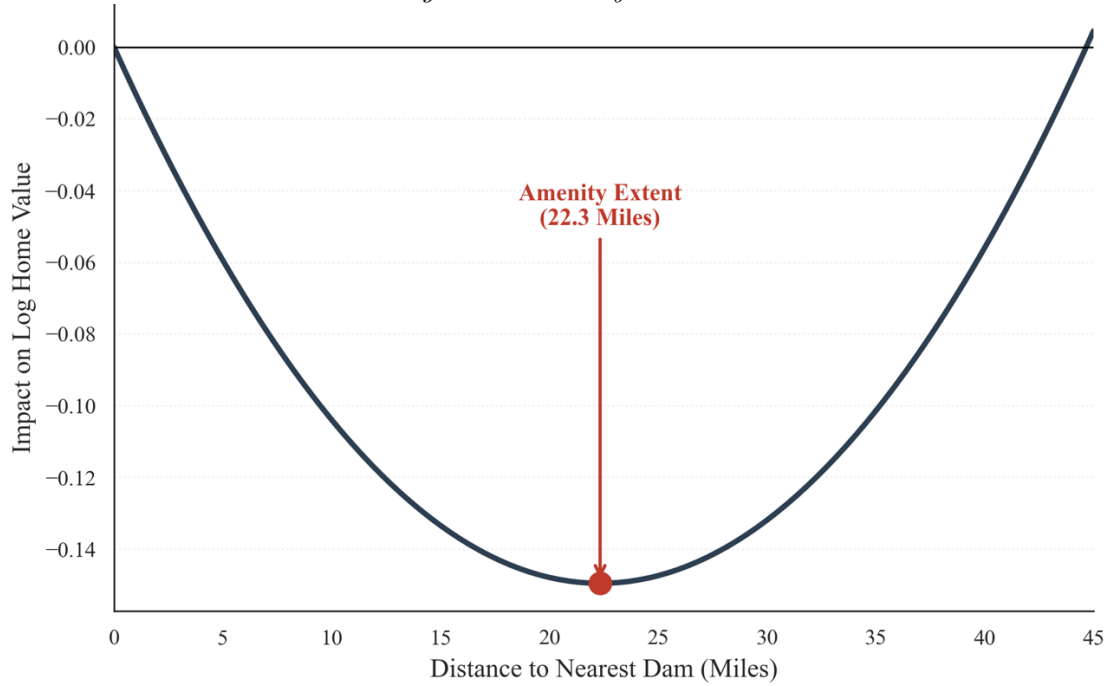
$$d^* = \frac{-\beta_1}{2\beta_2} = \frac{-(-0.0134)}{2(0.0003)} \approx 22.3 \text{ miles}$$

This result, visualized in Figure 3, suggests that the “Amenity Zone” for dams in North Carolina extends approximately 22 miles. Homes within this radius

* Notes: Dependent variable is Log Median Home Value. Standard errors are Heteroscedasticity-Consistent (HC3) and are reported in parentheses. Model 3 uses a Control Function approach where ‘1st Stage Resid.’ tests the endogeneity of Population Density; its significance ($p < 0.01$) rejects the null of exogeneity. $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

command a premium that decays with distance. Beyond this point, the influence of the dam becomes negligible or is overtaken by other regional factors.

Figure 3: “Amenity Zone” Distance



The vertical axis of Figure 3, labeled “Impact on Log Home Value,” quantifies the Total Price Contribution of the distance variables. The curve's starting point is negative ($\approx -1.34\%$), which indicates the dam is capitalized as an amenity. Specifically, a negative value on the Y-axis means that moving farther away from the dam causes the price to drop. Conversely, staying close commands a price premium. The upward slope of the line (becoming less negative) demonstrates that this decay rate is slowing. Critically, the curve reaches its most negative value—the point of maximum price decay—at the 22.3-mile threshold. This minimum point is the distance at which the marginal implicit price of proximity to the dam becomes zero and essentially signifies that the value of the amenity is fully dissipated and its influence on price stabilizes.

V. Conclusion

This study provides empirical evidence that dams in North Carolina generate positive externalities for local housing markets. By utilizing a Control Function approach where an Instrument Variable is used to correct for residential sorting, I identify a distinct “Amenity Zone” extending 22.3 miles from the dam’s infrastructure. These findings highlight the importance of accounting for non-linear spatial gradients when valuing public infrastructure and further support the use of hedonic pricing analysis to accurately measure the effects environmental factors play in housing markets.

The significance of the 22-mile threshold suggests that the recreational and aesthetic benefits of reservoirs are not merely local phenomena, but regional drivers of value. For policymakers, this implies that the economic benefits of dam maintenance and reservoir preservation extend across broader tax bases than previously estimated. Future research could refine these estimates by utilizing individual parcel data to disentangle the immediate disamenity risks of flood zones from the broader recreational benefits captured here.

A. Contribution to Hedonic Pricing Theory

These findings contribute to the evolving literature on non-market valuation by demonstrating the limitations of linear hedonic specifications. Traditional models often assume a constant marginal implicit price for environmental attributes (Rosen, 1974). My results, however, support the “Spatial Heterogeneity” hypothesis, which posits that the value of an amenity is structurally different depending on proximity. By identifying a specific turning point ($d^* \approx 22.3$ miles), this study provides a methodological template for defining the “market extent” of large-scale infrastructure, challenging arbitrary boundaries often used in policy analysis (e.g., fixed 5-mile radius).

B. Residential Sorting and Endogeneity

The divergence between the OLS and IV estimates underscores the critical role of Tiebout sorting in hedonic markets. The fact that high amenity areas attract higher population density—which itself influences price—creates a simultaneity

bias that masks the true value of the environmental good. This finding aligns with recent advances in urban economics (e.g., Kuminoff et al., 2013) that emphasize the need to treat location choice as an endogenous decision. Practically, this suggests that hedonic studies which fail to account for sorting likely underestimate the welfare benefits of public goods.

C. Policy Implications for Infrastructure Valuation

From a policy perspective, the existence of a wide “Amenity Zone” has significant implications for cost-benefit analysis and tax increment financing (TIF). If the capitalization of dam benefits extends over 20 miles, the tax base available to support infrastructure maintenance is considerably larger than the properties that reside in the dam’s immediate vicinity. This suggests that models for dam rehabilitation in which potential dam beneficiaries are responsible for the cost, might be expanded to encompass a larger geographic swath, rather than burdening only the most proximate residents. Furthermore, it highlights the dual nature of resilient infrastructure; dams are not merely risk management tools but contributors to regional wealth and housing stability.

D. Avenues for Future Research

The successful identification of a non-linear spatial gradient and the correction for residential sorting in this study open several critical paths for future research that can refine both empirical estimates and methodological understanding.

1. Disentangling Disamenity Risk from Amenity Benefits

While this study identified a clear 22.3-mile amenity threshold, future work needs to isolate the specific disamenity risks associated with dam proximity. Transitioning the analysis from aggregate ZCTA data to individual parcel level transactions would allow researchers to directly incorporate high-resolution variables. Specifically, including a variable for whether a property falls within the FEMA designated 100-year flood plain can separate the negative price effect of flood insurance risk from the positive aesthetic/recreational benefits of the reservoir, providing a cleaner causal estimate of the net impact. Future models should test the impact of secondary disamenities like noise pollution from dam operations, or

the visual obstruction created by the physical infrastructure itself, rather than relying solely on distance.

2. Heterogeneity in Infrastructure Characteristics

This project treated all dams as homogeneous infrastructure. However, the value capitalized may depend heavily on the dam's size, operational purpose, or age. Future hedonic models should introduce dummy variables or interaction terms for dam type (e.g., Hydroelectric vs. Flood Control), reservoir size (surface area), and public access to test for heterogeneity. This would determine if the amenity zone identified here is stable across different types of public goods or if the capitalization effect is conditional on specific infrastructure characteristics. This would provide richer policy guidance on which types of infrastructure generate the largest welfare benefits.

3. Testing the Robustness of the Instrument Variable (IV)

The correction for residential sorting relied on Log_Land_Area as an instrumental variable (IV). A crucial avenue for robustness is testing the sensitivity of the results to alternative instruments. Researchers could explore instruments that are highly correlated with population density but exogenous to local amenities, such as historic road network density or preindustrial geological features that constrain development capacity. This sensitivity analysis of the IV choice would further strengthen the causal claims regarding the impact of endogenous residential sorting on hedonic price estimates.

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Additional Figures and Tables

The following images illustrate the concepts outlining the study's hypothesis, diagnostic visualizations, and regression results supporting the empirical analysis of dam proximity and housing values in North Carolina. Most of these figures/tables were referenced in the paper and provide the foundational econometric analysis of my research, while others have been added as supplemental information.

Figures 5a and 5b detail the geographic region for which the analysis was sampled (NC). Both figures present a foundation which is comprised of all ZCTAs measured in this paper and illustrate levels of median home values and the locations of dam infrastructure, respectively. To validate the model specification, Figure 6 (Residuals vs. Fitted Values) is presented to assess potential heteroscedasticity. The graphical representation confirms the necessity of employing HC3 robust standard errors.

Figure 4: Home Values and Distance Distributions

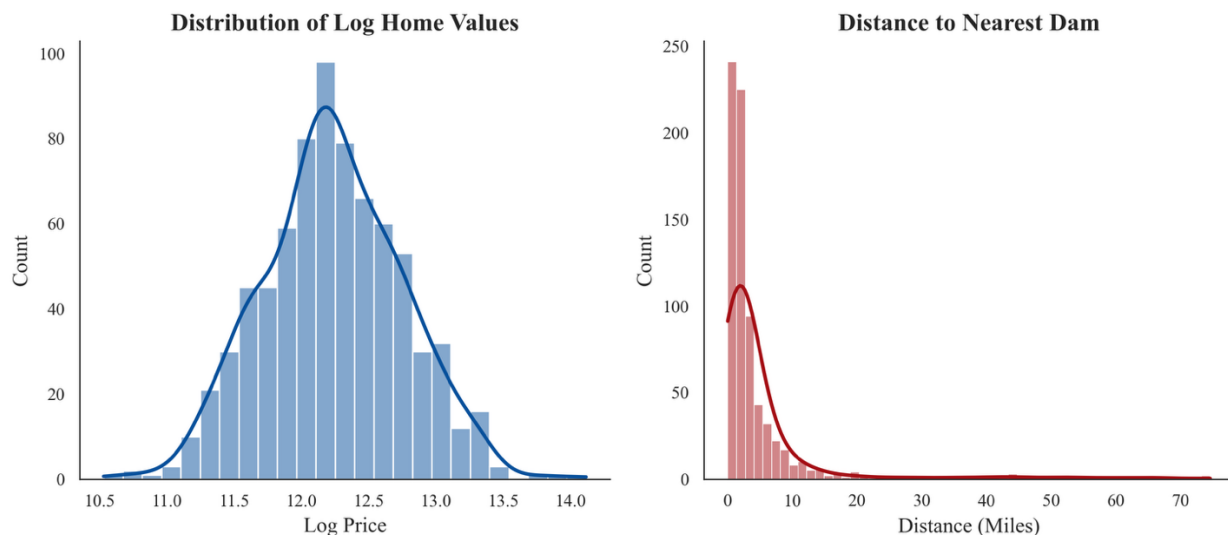


Figure 4. Histograms showing the frequency distribution of Log Median Home Values (left) and Distance to Nearest Dam across North Carolina ZCTAs (right). The bell-shaped curve in the left graph suggests the log-transformation successfully normalized the dependent variable.

Figure 5a: Spatial Distribution of Housing Density in North Carolina

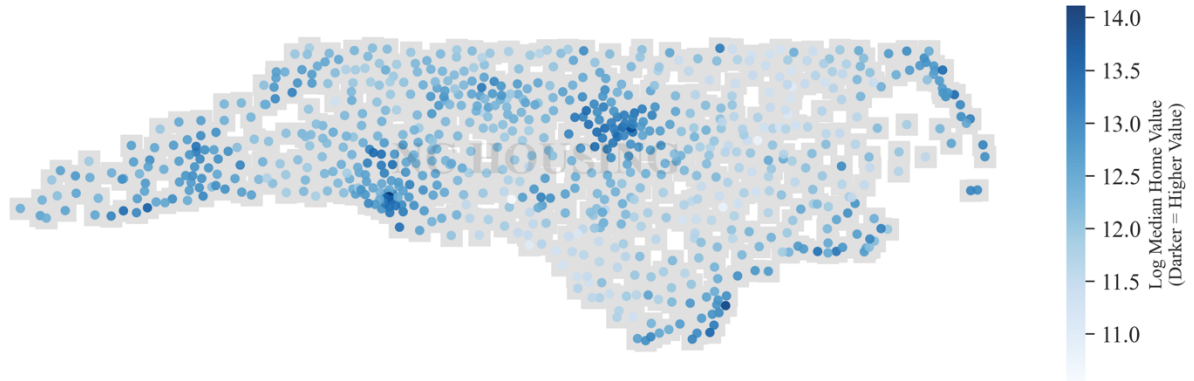


Figure 5b: Dam Infrastructure across North Carolina

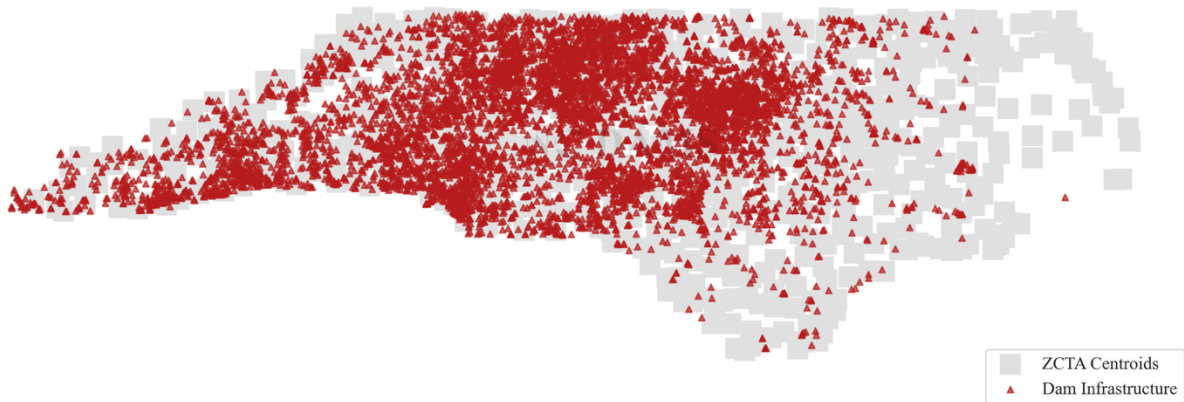


Figure 5a and 5b: Each circle in Figure 5a represents a distinct North Carolina ZCTA centroid within the study sample ($N = 747$). The shading indicates the median home price, where lighter blue represents lower values and darker blue represents higher values. Figure 5b displays the spatial distribution of dam infrastructure (red triangles) across the state. The base layer in both maps depicts the geographic footprint of all ZCTAs within the state to illustrate the complete study area.

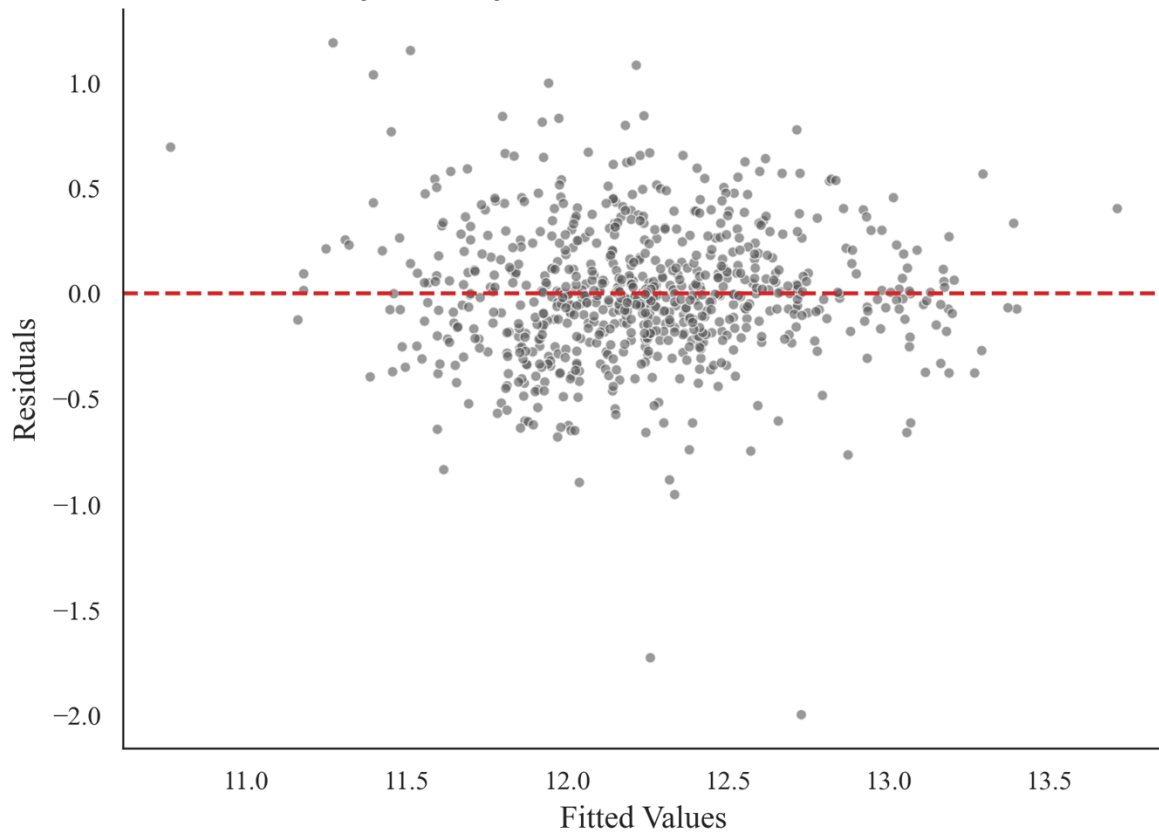
Figure 6: Diagnostic Residuals vs. Fitted Values

Figure 6: The random distribution of residuals confirms that the model is well-specified and that the significant findings regarding the amenity zone are not results of model misspecification or heteroscedasticity.

Appendix A: Replication Software and Code

All data processing and econometric analysis was performed using Python (v3.13). The primary libraries utilized include *pandas* for data manipulation, *statsmodels* for OLS estimation, *linearmodels* for instrumental variable analysis, and *matplotlib/seaborn* for visualization. Standard errors for all OLS specifications were calculated using the Heteroscedasticity-Consistent (HC3) covariance matrix. The below snippets of code were used in compiling different aspects of research and analysis for this paper.

A.1. Data Preparation and Variable Construction

This section details the transformation of raw Census and NID data into the final analytical dataset, including the calculation of the instrument.

```
df_final['Log_Home_Value']      =      np.log(df_final['Median_Home_Value'])

df_final['Distance_mi']      =      df_final['Distance_km']      *
KM_TO_MILES
df_final['Distance_SQ']      =      df_final['Distance_mi'] ** 2

df_final['ALAND_SqMi']      =      df_final['ALAND']      *
SQM_TO_SQMI
df_final['Population_Density']      =      df_final['Total_Population'] / df_final['ALAND_SqMi']
df_final['Log_Population_Density']      =      np.log(df_final['Population_Density'])
final_cols      =      ['ZCTA_ID',      'Log_Home_Value',      'Distance_mi', 'Distance_SQ', 'Median_Income', 'Log_Population_Density']
```

A.2. Geospatial Distance Calculation

The minimum distance from each ZCTA centroid to the nearest dam was calculated across all North Carolina .

```
min_distances_miles = []
for i, (zip_lat, zip_lon) in enumerate(zip_coords):
    nearest_dam_idx = indices[i]
    dam_lat = dam_coords[nearest_dam_idx][0]
    dam_lon = dam_coords[nearest_dam_idx][1]
    dist = geodesic((zip_lat, zip_lon), (dam_lat,
dam_lon)).miles
```



```

min_distances_miles.append(dist)

df_nc['Distance'] = min_distances_miles
df_nc['Distance_SQ'] = df_nc['Distance'] ** 2

```

A.3. OLS Estimation (Models 1 & 2)

Implementation of the Hedonic Price Function using robust standard errors.

```

m1_f = "Log_Home_Value ~ Nearest_Dam_Dist_mi + Dis-
tance_SQ + Log_Median_Income + Log_Population_Density"
model1 = smf.ols(m1_f, data=df).fit(cov_type='HC3')
m2_f = m1_f + " + Log_Median_Rooms"
model2 = smf.ols(m2_f, data=df).fit(cov_type='HC3')

```

A.4. Instrumental Variable (2SLS) and Endogeneity Testing

Implementation of the Two-Stage Least Squares strategy to test for simultaneity bias between Population Density and Housing Prices.

```

iv_formula = 'Log_Home_Value ~ 1 + Distance_mi + Dis-
tance_SQ + Log_Median_Income + Log_Median_Rooms +
[Log_Population_Density ~ Log_Land_Area]'
control_func_formula = 'Log_Home_Value ~ Log_Popula-
tion_Density + resid + Distance_mi + Distance_SQ +
Log_Median_Income + Log_Median_Rooms'
results_secstg = smf.ols(formula=control_func_formula,
data=df).fit(cov_type='HC3')
print(results_secstg.summary())

```

A.5. Generation of Figures

Code used to generate the distributional histograms and the quadratic fit scatter-plot.

```

plt.xlabel('Distance (Miles)', fontproperties=prop_la-
bel)
plt.ylabel('Frequency', fontproperties=prop_label)
ax = plt.gca()
for label in ax.get_xticklabels() + ax.get_ytick-
labels():
    label.set_fontproperties(prop_tick)
plt.tight_layout()
plt.savefig('Figure_1b_Distance.png', dpi=300)
print("Saved Figure_1b_Distance.png")
plt.show()

```